

AICE3001/AICE6002: Mathematics of Machine Learning

Subsidiary Notes — Lecture 8: Principal Component Analysis (PCA)

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1 Keywords

- Covariance matrices, positive semi-definite, dimensionality reduction, PCA, reconstruction error, duality

2 Main Points

2.1 Why Reduce Dimensions?

- Data often varies significantly in only a few directions, even when it has many features. You cannot usually see this by looking at the numbers
- Projecting onto the few directions of greatest variation gives fewer, more informative features for a learning machine, and discards directions that are mostly noise

2.2 Notation

- We have m data points $\mathbf{x}_1, \dots, \mathbf{x}_m$, each with p features, and sample mean $\boldsymbol{\mu} = \frac{1}{m} \sum_k \mathbf{x}_k$
- In this lecture we collect the *centred* data as the columns of a $p \times m$ matrix

$$\mathbf{X} = \frac{1}{\sqrt{m-1}} (\mathbf{x}_1 - \boldsymbol{\mu}, \mathbf{x}_2 - \boldsymbol{\mu}, \dots, \mathbf{x}_m - \boldsymbol{\mu})$$

This is the transpose of the design matrix used in linear regression (one row per data point), and it includes the factor $1/\sqrt{m-1}$

2.3 The Direction of Maximum Variation

- The spread of the data along a unit vector $\mathbf{v}$ ($\|\mathbf{v}\| = 1$) is the variance of the projections $\mathbf{v}^\top(\mathbf{x}_k - \boldsymbol{\mu})$

$$\sigma^2(\mathbf{v}) = \frac{1}{m-1} \sum_{k=1}^m (\mathbf{v}^\top(\mathbf{x}_k - \boldsymbol{\mu}))^2 = \mathbf{v}^\top \mathbf{C} \mathbf{v}$$

where, writing $(\mathbf{v}^\top \mathbf{a})^2 = \mathbf{v}^\top \mathbf{a} \mathbf{a}^\top \mathbf{v}$ and taking $\mathbf{v}$ outside the sum, $\mathbf{C}$ is the *covariance matrix*

$$\mathbf{C} = \frac{1}{m-1} \sum_{k=1}^m (\mathbf{x}_k - \boldsymbol{\mu})(\mathbf{x}_k - \boldsymbol{\mu})^\top = \mathbf{X} \mathbf{X}^\top$$

- To maximise $\mathbf{v}^\top \mathbf{C} \mathbf{v}$ subject to $\mathbf{v}^\top \mathbf{v} = 1$ we use a Lagrange multiplier. Setting the gradient of $\mathcal{L} = \mathbf{v}^\top \mathbf{C} \mathbf{v} - \lambda(\mathbf{v}^\top \mathbf{v} - 1)$ to zero gives

$$2 \mathbf{C} \mathbf{v} - 2 \lambda \mathbf{v} = \mathbf{0} \quad \Rightarrow \quad \mathbf{C} \mathbf{v} = \lambda \mathbf{v}$$

(we used $\nabla \mathbf{v}^\top \mathbf{C} \mathbf{v} = 2 \mathbf{C} \mathbf{v}$ for symmetric $\mathbf{C}$, from the mappings lecture)

- So the stationary directions are the eigenvectors of $\mathbf{C}$, and the variance along eigenvector $\mathbf{v}_i$ is $\mathbf{v}_i^T \mathbf{C} \mathbf{v}_i = \lambda_i \mathbf{v}_i^T \mathbf{v}_i = \lambda_i$. The direction of greatest variation is the eigenvector with the largest eigenvalue
- The next best direction *orthogonal to the first* is the eigenvector with the second largest eigenvalue, and so on

2.4 Properties of the Covariance Matrix

- $C_{ij} = \frac{1}{m-1} \sum_k (x_{ik} - \mu_i)(x_{jk} - \mu_j)$ measures how features i and j vary together, the sample version of $\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$. The diagonal holds the variances of the individual features
- $\mathbf{C}$ is built from *outer products* $\mathbf{a} \mathbf{a}^T$ (a $p \times p$ matrix), compared with the inner product $\mathbf{a}^T \mathbf{a}$ (a number)
- $\mathbf{C}$ is symmetric, so its eigenvectors can be chosen orthonormal (eigensystems lecture)
- $\mathbf{C}$ is *positive semi-definite*: for any $\mathbf{v}$,

$$\mathbf{v}^T \mathbf{C} \mathbf{v} = \mathbf{v}^T \mathbf{X} \mathbf{X}^T \mathbf{v} = \|\mathbf{X}^T \mathbf{v}\|^2 \geq 0$$

Applying this to an eigenvector shows every eigenvalue is $\lambda_i = \mathbf{v}_i^T \mathbf{C} \mathbf{v}_i / \|\mathbf{v}_i\|^2 \geq 0$ (they are variances, after all)

- The total variance, summed over all features, is $\text{Tr} \mathbf{C} = \sum_i C_{ii} = \sum_i \lambda_i$. The fraction $\sum_{i \leq k} \lambda_i / \sum_i \lambda_i$ is the "variance explained" by the first k components
- **Zero eigenvalues.** $\lambda_i = 0$ exactly when there is no variation at all along $\mathbf{v}_i$
 - The m centred points sum to $\mathbf{0}$, so they span at most an $(m-1)$ -dimensional subspace. If $m \leq p$, $\mathbf{C}$ must have zero eigenvalues
 - With $m \geq p+1$ points that do not all lie in one hyperplane, every eigenvalue is positive: $\mathbf{C}$ is *positive definite*, full rank and invertible
 - Typically, with $m > p$, $\mathbf{C}$ is positive definite unless the features are linearly related (e.g. one is the sum of others)
- **The ellipsoid picture.** The surface $\mathbf{x}^T \mathbf{C}^{-1} \mathbf{x} = 1$ is an ellipsoid whose principal axes point along the eigenvectors. Along $\mathbf{v}_i$, putting $\mathbf{x} = r \mathbf{v}_i$ gives $r^2 / \lambda_i = 1$, so the radius is $\sqrt{\lambda_i}$: the standard deviation of the data in that direction

2.5 Principal Component Analysis

- The recipe
 1. Compute the mean $\boldsymbol{\mu}$ and the covariance matrix $\mathbf{C}$
 2. Find its eigenvalues and eigenvectors
 3. Keep the k eigenvectors with the largest eigenvalues, the *principal components*
 4. Project each input into the space they span
- The projection uses the $k \times p$ matrix whose rows are the principal components

$$\mathbf{P} = \begin{pmatrix} \mathbf{v}_1^T \\ \vdots \\ \mathbf{v}_k^T \end{pmatrix}, \quad \mathbf{z} = \mathbf{P} (\mathbf{x} - \boldsymbol{\mu})$$

The k numbers $z_j = \mathbf{v}_j^T (\mathbf{x} - \boldsymbol{\mu})$ are the new features

- **Reconstruction.** We can map back to an approximation of the input

$$\hat{\mathbf{x}} = \boldsymbol{\mu} + \sum_{j=1}^k z_j \mathbf{v}_j$$

- **Reconstruction error.** All p eigenvectors form an orthonormal basis, so $\mathbf{x} - \boldsymbol{\mu} = \sum_{j=1}^p z_j \mathbf{v}_j$ exactly, and the error is the part we dropped, $\mathbf{x} - \hat{\mathbf{x}} = \sum_{j>k} z_j \mathbf{v}_j$. By orthonormality $\|\mathbf{x} - \hat{\mathbf{x}}\|^2 = \sum_{j>k} z_j^2$, and averaging over the data (with the same $1/(m-1)$) each z_j^2 averages to the variance along $\mathbf{v}_j$, which is λ_j . So

$$\frac{1}{m-1} \sum_{k'=1}^m \|\mathbf{x}_{k'} - \hat{\mathbf{x}}_{k'}\|^2 = \sum_{j>k} \lambda_j$$

the sum of the eigenvalues we ignored. No other k -dimensional subspace does better, which is why PCA keeps the largest eigenvalues

- **Choosing k .** Plot the eigenvalues in decreasing order (as for the handwritten digits in the lecture) and keep enough components to explain most of the variance. There is no single right answer
- Images of handwritten digits: the leading eigenvectors look like blurred digits, and a few tens of them reconstruct a recognisable digit from its hundreds of pixels

2.6 Duality

- For images p is huge: a 256×256 image has $p = 65\,536$ pixels, and $\mathbf{C}$ would have over 4×10^9 entries. But m images span at most an $(m-1)$ -dimensional subspace, usually with $m \ll p$
- Work instead with the $m \times m$ *dual matrix* $\mathbf{D} = \mathbf{X}^T \mathbf{X}$, whose entries $D_{kl} = \frac{1}{m-1} (\mathbf{x}_k - \boldsymbol{\mu})^T (\mathbf{x}_l - \boldsymbol{\mu})$ are inner products between data points. Building it costs $O(p m^2)$
- If $\mathbf{D} \mathbf{v} = \lambda \mathbf{v}$, multiply on the left by $\mathbf{X}$:

$$\mathbf{X} \mathbf{X}^T \mathbf{X} \mathbf{v} = \lambda \mathbf{X} \mathbf{v} \quad \Rightarrow \quad \mathbf{C} \mathbf{u} = \lambda \mathbf{u}, \quad \mathbf{u} = \mathbf{X} \mathbf{v}$$

so every eigenvector of $\mathbf{D}$ with $\lambda \neq 0$ gives one of $\mathbf{C}$ with the same eigenvalue. (If $\lambda = 0$ then $\mathbf{X} \mathbf{v} = \mathbf{0}$ and there is nothing to map)

- $\mathbf{C}$ and $\mathbf{D}$ therefore have the same *non-zero* eigenvalues. The bigger matrix just has extra zeros
- **Normalising.** $\|\mathbf{X} \mathbf{v}\|^2 = \mathbf{v}^T \mathbf{D} \mathbf{v} = \lambda$ for a unit $\mathbf{v}$, so the unit eigenvector of $\mathbf{C}$ is $\mathbf{u} = \mathbf{X} \mathbf{v} / \sqrt{\lambda}$
- We have worked in the *dual space*, spanned by the examples. The same trick reappears in SVMs and kernel methods, and the pair $\mathbf{X} \mathbf{X}^T$, $\mathbf{X}^T \mathbf{X}$ is the heart of the SVD (next lecture)

3 Exercises

1. Show that $\frac{1}{m-1} \sum_k (\mathbf{v}^T (\mathbf{x}_k - \boldsymbol{\mu}))^2 = \mathbf{v}^T \mathbf{C} \mathbf{v}$
2. Two data points $\mathbf{y}^1 = (2, 4, 5)^T$ and $\mathbf{y}^2 = (8, 6, 1)^T$ (the example in the lecture)
 - (a) Find $\boldsymbol{\mu}$ and $\mathbf{X}$ (with $m = 2$ the factor $1/\sqrt{m-1}$ is 1)
 - (b) Find $\mathbf{D} = \mathbf{X}^T \mathbf{X}$ and its eigenvalues and eigenvectors
 - (c) Use the dual to find the principal component of $\mathbf{C}$, and explain why $\mathbf{C}$ has two zero eigenvalues

3. Show that $\text{Tr } \mathbf{C}$ equals the sum of the variances of the individual features, and also the sum of the eigenvalues
4. Data in two dimensions have covariance matrix $\mathbf{C} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$
 - (a) Find the principal components and the variance along each
 - (b) What fraction of the variance does the first component explain?
 - (c) What are the radii and directions of the axes of the ellipse $\mathbf{x}^T \mathbf{C}^{-1} \mathbf{x} = 1$?
5. Show that if $\mathbf{v}$ is a unit eigenvector of $\mathbf{D}$ with eigenvalue $\lambda > 0$ then $\mathbf{X} \mathbf{v} / \sqrt{\lambda}$ is a unit eigenvector of $\mathbf{C}$

4 Experiments

4.1 PCA by hand

- Generate data in 10 dimensions that really lies close to a plane, and check the claims in these notes

```
import numpy as np

rng = np.random.default_rng(0)
m, p = 200, 10
basis = rng.normal(size=(p, 2)) # data really lives near a plane
Y = (basis @ rng.normal(size=(2, m))).T + 0.1 * rng.normal(size=(m, p))
mu = Y.mean(axis=0)
X = (Y - mu).T / np.sqrt(m - 1) # p x m, columns are centred points
C = X @ X.T # covariance matrix (p x p)
lam, V = np.linalg.eigh(C)
lam, V = lam[::-1], V[:, ::-1] # largest eigenvalue first
print(np.round(lam, 3)) # two big, eight tiny
print(np.allclose(C, np.cov(Y.T))) # agrees with numpy's covariance

for k in (1, 2, 3):
    P = V[:, :k].T # k x p projection matrix
    Z = P @ (Y - mu).T # k x m new features
    Yhat = mu + (V[:, :k] @ Z).T # reconstruction
    err = np.sum((Y - Yhat) ** 2) / (m - 1)
    # the error should equal the sum of the discarded eigenvalues
    print(k, round(err, 4), round(lam[k:].sum(), 4))

# duality: with more features than examples use the m x m matrix
Xs = X[:, :5] # pretend we have only 5 examples
print(np.round(np.linalg.eigvalsh(Xs.T @ Xs)[::-1], 4)) # 5 eigenvalues of D
print(np.round(np.linalg.eigvalsh(Xs @ Xs.T)[::-1], 4)) # same, plus zeros
```

- Two eigenvalues (about 12 and 2.7) stand out; the other eight are around 0.01, the noise. Keeping $k = 2$ components cuts the reconstruction error from about 2.8 to 0.08, and in every case the error equals the sum of the discarded eigenvalues
- The 5×5 dual matrix has the same non-zero eigenvalues as the 10×10 matrix, which just has five extra zeros

5 Answers

1. $(\mathbf{v}^\top \mathbf{a})^2 = (\mathbf{v}^\top \mathbf{a})(\mathbf{a}^\top \mathbf{v}) = \mathbf{v}^\top \mathbf{a} \mathbf{a}^\top \mathbf{v}$, since $\mathbf{v}^\top \mathbf{a}$ is a number and equals its transpose. With $\mathbf{a} = \mathbf{x}_k - \boldsymbol{\mu}$, summing over k and taking $\mathbf{v}$ outside the sum gives $\mathbf{v}^\top \mathbf{C} \mathbf{v}$

2. The lecture example

(a) $\boldsymbol{\mu} = (5, 5, 3)^\top$ and $\mathbf{X} = \begin{pmatrix} -3 & 3 \\ -1 & 1 \\ 2 & -2 \end{pmatrix}$

(b) $\mathbf{D} = \begin{pmatrix} 14 & -14 \\ -14 & 14 \end{pmatrix}$, with $\lambda = 28$, $\mathbf{v} = (-1, 1)^\top / \sqrt{2}$ and $\lambda = 0$, $\mathbf{v} = (1, 1)^\top / \sqrt{2}$

(c) $\mathbf{X} \mathbf{v} = (6, 2, -4)^\top / \sqrt{2}$ and dividing by $\sqrt{28}$ gives $\mathbf{u} = (3, 1, -2)^\top / \sqrt{14}$, with variance 28. The two centred points are $\pm(3, 1, -2)^\top$: they lie on a line through the origin, so there is no variation in the two directions perpendicular to it

3. $\text{Tr } \mathbf{C} = \sum_i C_{ii}$ and C_{ii} is the variance of feature i . Using $\mathbf{C} = \mathbf{V} \boldsymbol{\Lambda} \mathbf{V}^\top$ and the cyclic property of the trace, $\text{Tr } \mathbf{C} = \text{Tr } \boldsymbol{\Lambda} \mathbf{V}^\top \mathbf{V} = \text{Tr } \boldsymbol{\Lambda} = \sum_i \lambda_i$

4. The matrix from the eigensystems notes

(a) $\lambda = 3$ along $(1, 1)^\top / \sqrt{2}$ and $\lambda = 1$ along $(1, -1)^\top / \sqrt{2}$. The variances are 3 and 1

(b) $3/(3+1) = 75\%$

(c) Radii $\sqrt{3}$ along $(1, 1)^\top$ and 1 along $(1, -1)^\top$

5. $\mathbf{C}(\mathbf{X} \mathbf{v}) = \mathbf{X}(\mathbf{X}^\top \mathbf{X} \mathbf{v}) = \lambda \mathbf{X} \mathbf{v}$, so it is an eigenvector, and $\|\mathbf{X} \mathbf{v}\|^2 = \mathbf{v}^\top \mathbf{X}^\top \mathbf{X} \mathbf{v} = \lambda \mathbf{v}^\top \mathbf{v} = \lambda$, so dividing by $\sqrt{\lambda}$ makes it a unit vector